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ETW3482

Week Nine Tutorial



This week's tutorial:

- **Questions for Revision**
- **Hands-on activities with Quizzes**



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Game time!



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What does the interpretability of a predictive model primarily refer to?

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Which of the following is NOT a reason why interpretability is important for stakeholders?

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What is the main trade-off when improving a model's interpretability?

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Which of the following is an example of an inherently interpretable model?

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Methods for Model Interpretability

- Model interpretability encompasses two main categories: inherently interpretable models and post-hoc explanations.
- **Inherently interpretable models** are designed with transparency in mind from the outset, directly incorporating interpretability into their structure. Examples include generalised linear models, which offer straightforward explanations through their coefficient estimates.
- **Post-hoc explanations**, on the other hand, are techniques applied after a model is trained to explain its predictions. These methods aim to provide insights into the inner workings of complex models, making them more understandable to humans.

Global and Local Interpretability

- Within post-hoc interpretability, techniques can be broadly categorised into **global and local interpretability** methods.
- **Global interpretability techniques** strive to reveal the model's overall behaviour across the entire dataset. They reveal overarching patterns, such as which features most influence driving predictions or how each feature impacts the model's overall output.
- **Local interpretability methods**, conversely, zoom in on individual predictions, offering explanations tailored to specific instances. By focusing on individual cases, these techniques provide insights into why a particular prediction was made, offering valuable context and understanding.

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What does a Partial Dependence (PD) plot primarily show?

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Global Interpretability: Variable Importance

- Variable importance tables and partial dependence (PD) plots are two commonly used global interpretability techniques, and you can access them in Model Studio.
- Variable Importance Tables: These tables provide insight into the importance of each feature in the predictive model. By indicating the statistical contribution of each feature, stakeholders can understand which variables have the most influence on the model's predictions. Various methods exist for calculating feature importance, such as fitting a global surrogate decision tree model or using permutation-based feature importance.

Global Interpretability: PD Plots

- Partial Dependence Plots (PD): PD plots visually represent the relationship between a specific feature and the model's average prediction. By varying the values of one feature while keeping others constant and observing the resulting change in predictions, stakeholders can understand how a feature influences the model's output on average across the entire dataset.
- Refer to How to Explain Your Black-Box Models in SAS Viya for the steps to produce PD plots.

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Which local interpretability technique uses a surrogate linear model to explain predictions?

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Local Interpretability

- Local interpretability methods aim to explain individual predictions of machine learning models.
- The common techniques are Individual conditional expectation (ICE) plots, LIME (Local Interpretable Model-Agnostic Explanations) and Shapley Values.
- These techniques focus on explaining why a specific prediction was made for a particular instance.

Local Interpretability: LIME

- LIME (Local Interpretable Model-Agnostic Explanations) is a technique used to explain individual predictions of any machine-learning model by building a white-box local surrogate model.
- It works by sampling the feature space around a specific observation and then training a sparse linear regression model, such as LASSO, on this generated sample.
- The surrogate model approximates the behaviour of the black-box model locally, making it easier to understand.

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Which method fairly distributes the contribution of each feature to a model's prediction by considering all possible combinations of features?

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Local Interpretability: Shapley Values

- Shapley values are a method for explaining individual predictions by averaging across all permutations of the features.
- Unlike LIME coefficients, Shapley values control for variable interactions, making them more robust in capturing the contribution of each feature to the prediction.
- This allows for a more comprehensive understanding of how each feature contributes to the model's predictions.

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Which technique offers a computationally efficient way to compute Shapley values using synthetic data?

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Local Interpretability: Kernel SHAP

- Kernel SHAP is a method for computing Shapley values based on synthetic data generation and weighted regression.
- It provides an efficient way to compute Shapley values by generating synthetic data that follows the distributions of the input data set and then scoring this data using the machine learning model.
- Kernel SHAP is implemented in SAS Viya, offering users a computationally efficient approach to compute Shapley values.
- By leveraging synthetic data generation and weighted regression, Kernel SHAP provides accurate explanations for individual predictions, making it a valuable tool for model interpretability in SAS Viya.

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PART 2:
PRACTICAL HANDS-ON SESSION



Preparation for the practical session

1. Open the demo project.
2. Access the Week 8 tutorial pipeline.
3. Select the SVM Node.
4. Refer to the SVM node property panel.

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Preparation for the practical session

1. Refer to instructions No. 2 to 4 in the “Week 9 Demo Guide” to include the Global and Local Interpretability.
2. Run the SVM Node
3. Open the results for the node and click the Model Interpretability tab.

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Expand the Surrogate Model Variable Importance table in the Model Interpretability tab. What is the most important input?

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Explain the relationship between Handset Age Group variable and the model's prediction.

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Expand the PD Plot.
Explain the relationship between Total Days Over Plan (ever_days_over_plan) variable and the model's prediction.

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Expand the PD and ICE Overlay Plot.
Does the Total Days Over Plan variable have an interaction effect?

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Expand the LIME Explanations plot.
What is the most significant input for the local regression model for the individual observation that is displayed on the LIME plot?

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Why would you apply Kernel SHAP after building and tuning an SVM model like the one in the practical lesson?

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Suppose your tuned SVM with a polynomial kernel shows improved performance.
What can Kernel SHAP help you verify about that improvement?

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WRAP UP



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Q&A



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Audience Q&A

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